

FeastFleet

Intelligent Hyperlocal Food Delivery Platform
with AI Culinary Services

Software Architecture & Engineering
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Fill remaining names and roll numbers before final print submission if not already printed on this page.

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1. Introduction

1.1 Brief description of the software product

FeastFleet is a full-stack web application for ordering food from nearby restaurants, with a focus on Kerala / hyperlocal delivery. Users can browse restaurants, view menus, add dishes to a cart, place orders, and follow a live delivery timeline. The product also includes AI-assisted tools: a personal food agent, budget optimizer, food memory profile, visual food search, group-order planner, ingredient/allergen predictions, and FeastBot (a floating chatbot).

The frontend is a multi-page HTML/CSS/JavaScript interface. The backend is an Express.js REST API with JWT authentication and MongoDB (or an automatic in-memory MongoDB fallback for demos).

1.2 Purpose of the system

The system is designed to:

- Let customers discover restaurants and dishes and complete an order in a few steps.
- Show proximity (distance and estimated time) based on a set delivery location.
- Keep cart items consistent by limiting multi-restaurant carts to a 1 km radius.
- Help users choose food using budget, diet, group, and allergen constraints.
- Provide order history and live tracking after checkout.

1.3 Intended users

User type	How they use FeastFleet
Customers / end users	Register or sign in, search restaurants, order food, track delivery, contact support.

User type	How they use FeastFleet
Demo / evaluators	Use 1-click demo login, walk through AI tools, and inspect the running API.
Developers / testers	Install Node.js, start the server, and exercise frontend + REST endpoints.

2. System Requirements

2.1 Hardware requirements

Item	Minimum	Recommended
Processor	Dual-core CPU	Quad-core or better
RAM	4 GB	8 GB or more
Disk space	~300 MB for project + Node modules	1 GB free
Display	1366 × 768	1920 × 1080 (desktop)
Network	Localhost is enough for demo	Internet for CDN (Bootstrap, icons, some images)

2.2 Software requirements

- **Node.js** 18 or later (includes npm)
- **Optional:** MongoDB Community Server. If MongoDB is not installed, the server starts `mongodb-memory-server` automatically.
- Git (optional, for cloning the repository)

2.3 Browser / platform requirements

- **Supported browsers:** Google Chrome, Microsoft Edge, Firefox, or Safari (latest two versions).
- **OS:** Windows 10/11, macOS, or Linux.
- JavaScript must be enabled. A modern browser is required for geolocation UI, localStorage, and fetch APIs.
- **Default URL:** `http://localhost:5000` (port 5000 unless `PORT` is set in the environment).

3. Installation / Setup

These steps start FeastFleet on the local machine for demonstration or academic evaluation.

1. Install Node.js from `https://nodejs.org` and confirm in a terminal: `node -v` and `npm -v`.
2. Open the project folder (example): `feastfleet`.
3. Install dependencies: `npm install`.
4. Start the application: `npm start`.
5. Wait until the console prints that the FeastFleet server is running on port 5000.

6. Open a browser and go to `http://localhost:5000` (or `http://localhost:5000/index.html`).

Optional seed: `npm run seed` loads restaurant/menu data into MongoDB. The in-memory database used for demos is seeded as part of a normal start when configured that way; if the catalog looks empty, run seed and restart.

Access without a full install: HTML files can be opened directly, but login, AI APIs, and order sync need the Express server. Always use `npm start` for a complete demo.

On first load you should see the brand splash screen, then the home storefront.

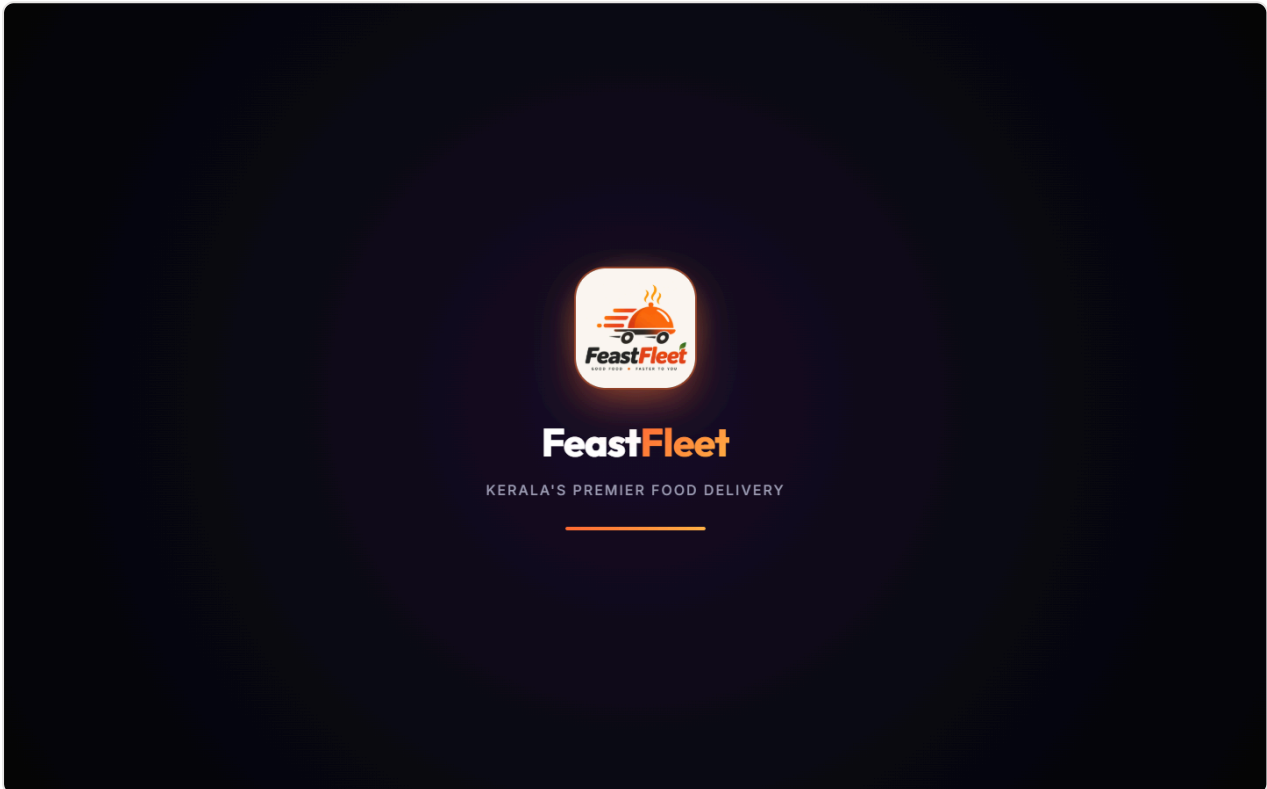


Figure 1. Splash screen shown for about 2.5 seconds at launch.

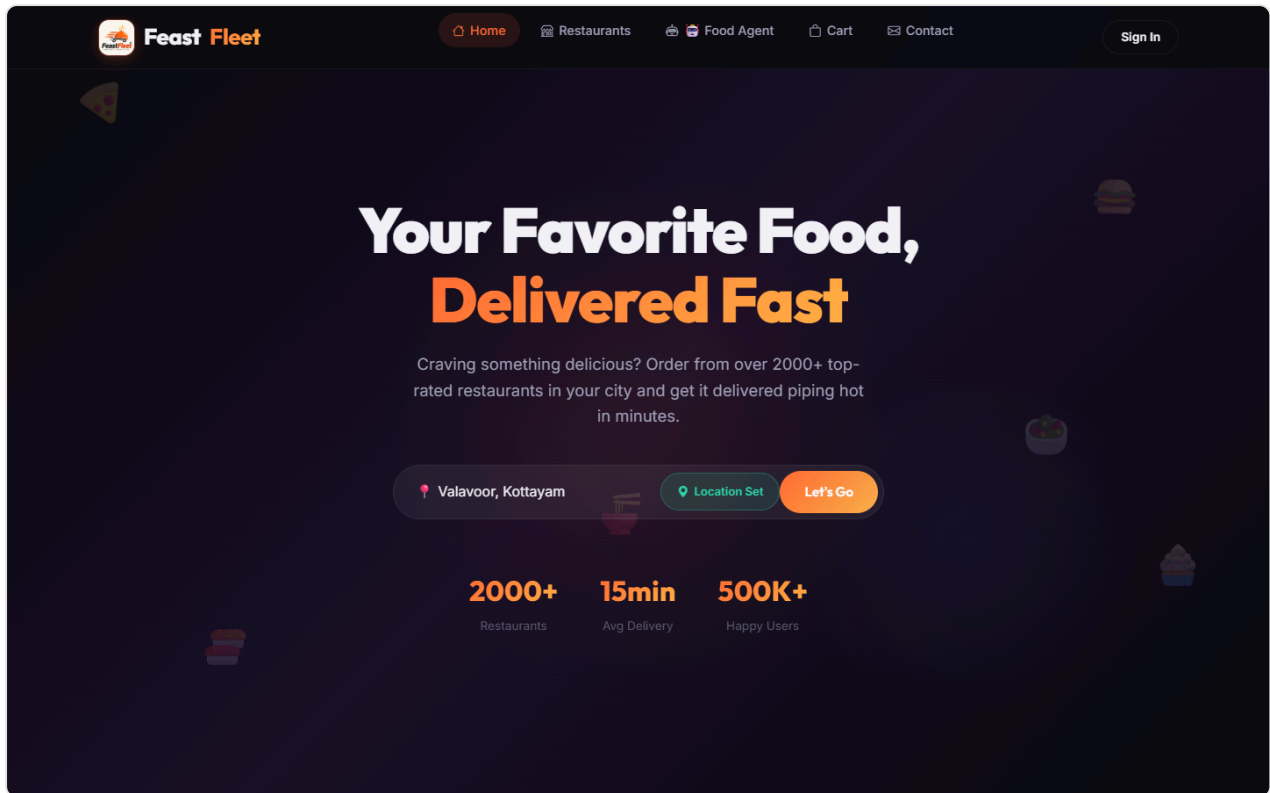


Figure 2. Home page after the splash animation.

4. User Login / Registration

Open **Sign In** from the top-right of the navbar, or go to `login.html` .

4.1 Sign in (existing account)

1. Keep the **Sign In** tab active.
2. Enter email, username, or mobile in **Email / Username / Mobile**.
3. Enter the password (minimum 6 characters on the server).
4. Click **Sign In**.
5. On success you see a welcome message and are redirected to Home. The navbar shows your initials instead of Sign In.

The image shows a web application interface for 'Feast Fleet'. At the top, there is a navigation bar with links for Home, Restaurants, Cart, and Contact, along with a 'Sign In' button. The main content area features a 'Welcome Back' sign-in form. The form includes a title 'Welcome Back', a subtitle 'Sign in with email, username or mobile', and two buttons: 'Sign In' (highlighted in orange) and 'Sign Up'. Below these are input fields for 'Email / Username / Mobile' (with placeholder text 'name@example.com, username or phone') and 'Password' (with masked characters '*****'). A large orange 'Sign In' button is positioned below the password field. At the bottom of the form, there is a link 'or quick demo sign in' and a button labeled '1-Click Demo Login (Abhinav)'.

Figure 3. Sign In form.

Demo login (recommended for review): click **1-Click Demo Login (Abhinav)**. Demo account:

- Name: Abhinav Babu
- Email: `abhinav@feastfleet.com`
- Password: `password123`

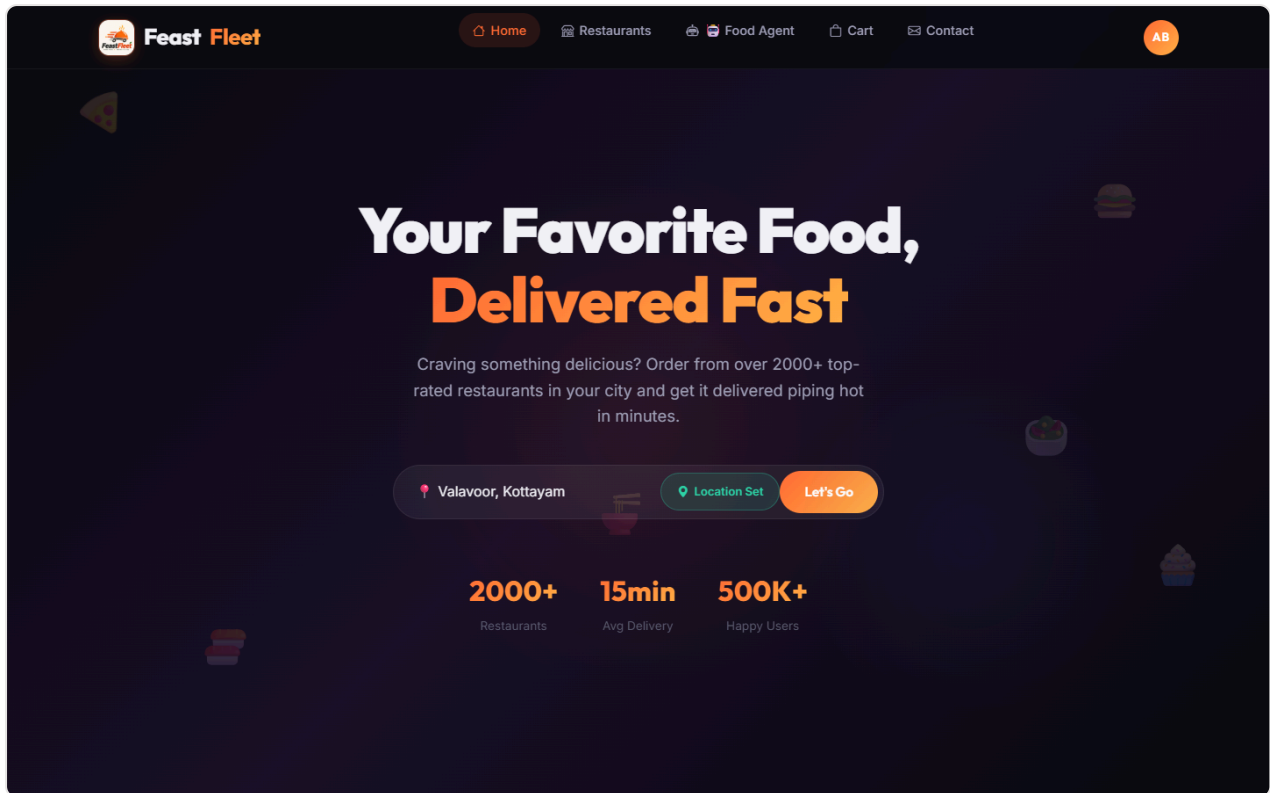


Figure 4. Home page after a successful sign-in (profile avatar on the right).

4.2 Registration (new account)

1. Click the **Sign Up** tab.
2. Enter **Full Name**, **Mobile Number (+91)**, **Email Address**, and **Password**.
3. Click **Create Account / Send OTP & Create Account**.
4. Enter any 4-digit OTP (demo mode). The account is created and you are signed in.

Join FeastFleet
Create your account and start ordering

Sign In Sign Up

Full Name
e.g. Abhinav Babu

Mobile Number
+91 9876543210

Email Address
name@example.com

Password

Create Account

or quick demo sign in

1-Click Demo Login (Abhinav)

Figure 5. Sign Up form.

If the API is unreachable, the page still signs you in locally so the rest of the demo can continue.

5. System Features / Functionalities

5.1 Set delivery location

1. On Home, use the address field or click **Detect My Location** / location control.
2. For this build, detection sets **Valavoor, Kottayam**.
3. Restaurant cards then show distance (km) and estimated time.

Expected result: location label updates; nearby restaurants (for example AK Take Away, Pala) appear closer than Kochi outlets.

5.2 Browse restaurants

1. Click **Restaurants** in the navbar, or **Let's Go** from the home search.
2. Use cuisine chips (All, Burgers, Asian, Indian / Kerala, Desserts, and similar) to filter.
3. Click a restaurant card to open its menu. AK Take Away, Pala opens a dedicated menu page.

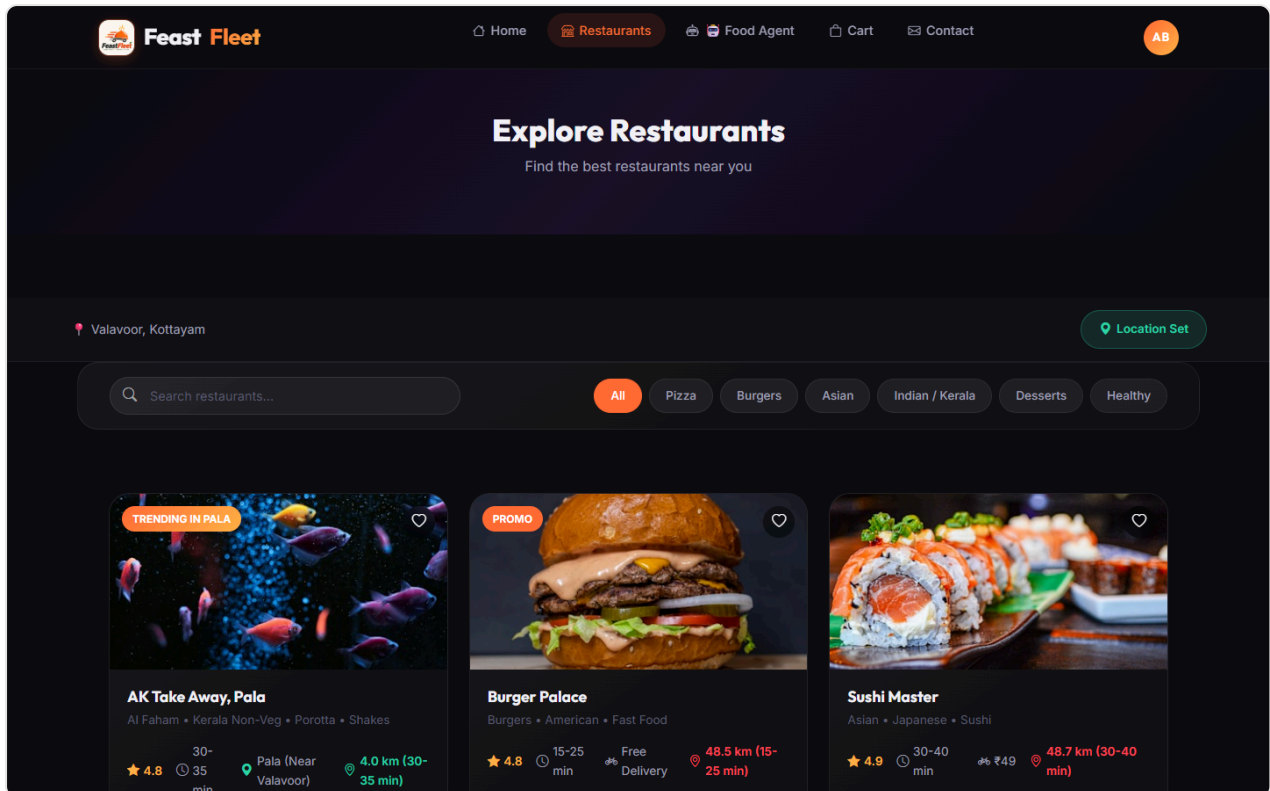


Figure 6. Restaurant catalog.

5.3 View menu and add items

1. Open a restaurant (example: AK Take Away, Pala).
2. Filter by category chips (Al Faham, Kerala Non-Veg, Vegetarian, Rice & Biryani, and so on).
3. Click the orange + button on a dish to add it to the cart.
4. The cart badge in the navbar increases.

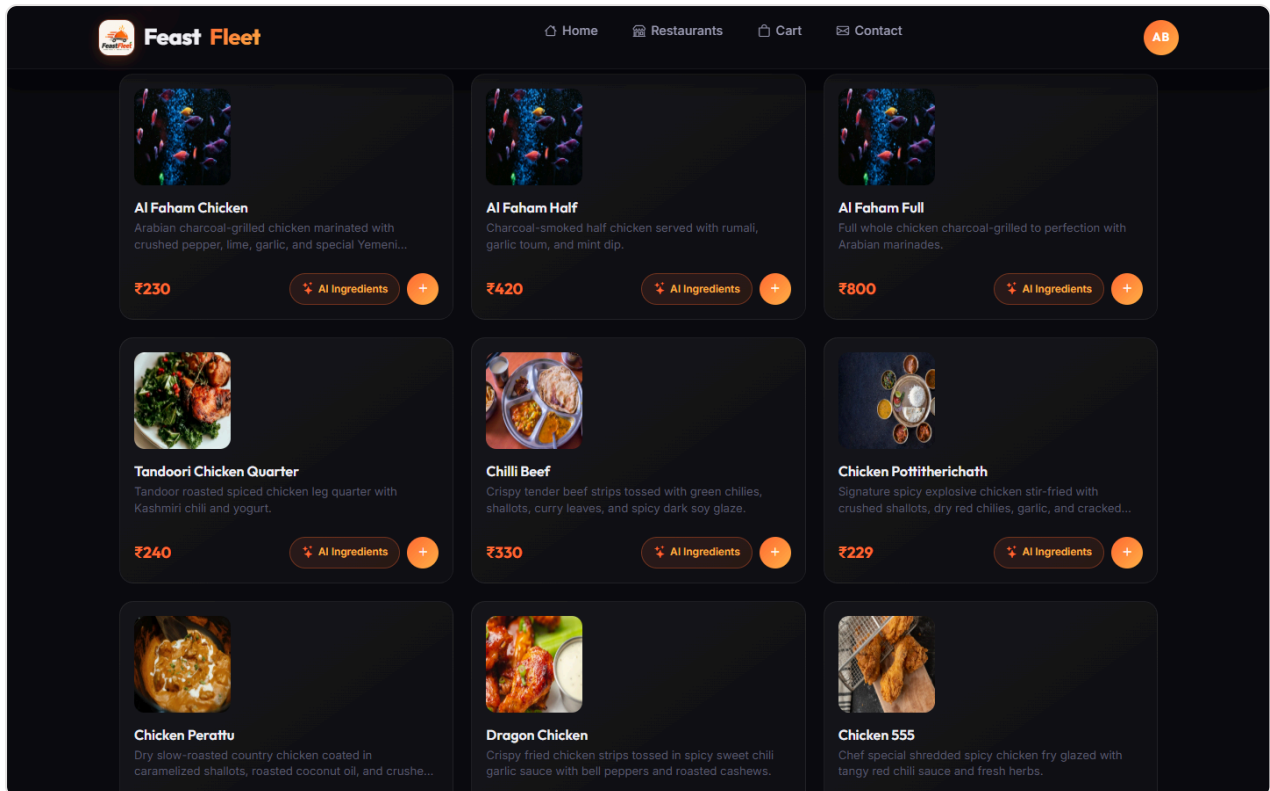


Figure 7. Menu with prices, AI Ingredients, and add-to-cart controls.

5.4 AI ingredient and allergen prediction

1. On a dish card, click **AI Ingredients**.
2. Read predicted ingredients, dietary notes, flavor notes, match %, and allergen warnings.
3. Optionally click **Ask AI Chatbot**, or **Done** to close.

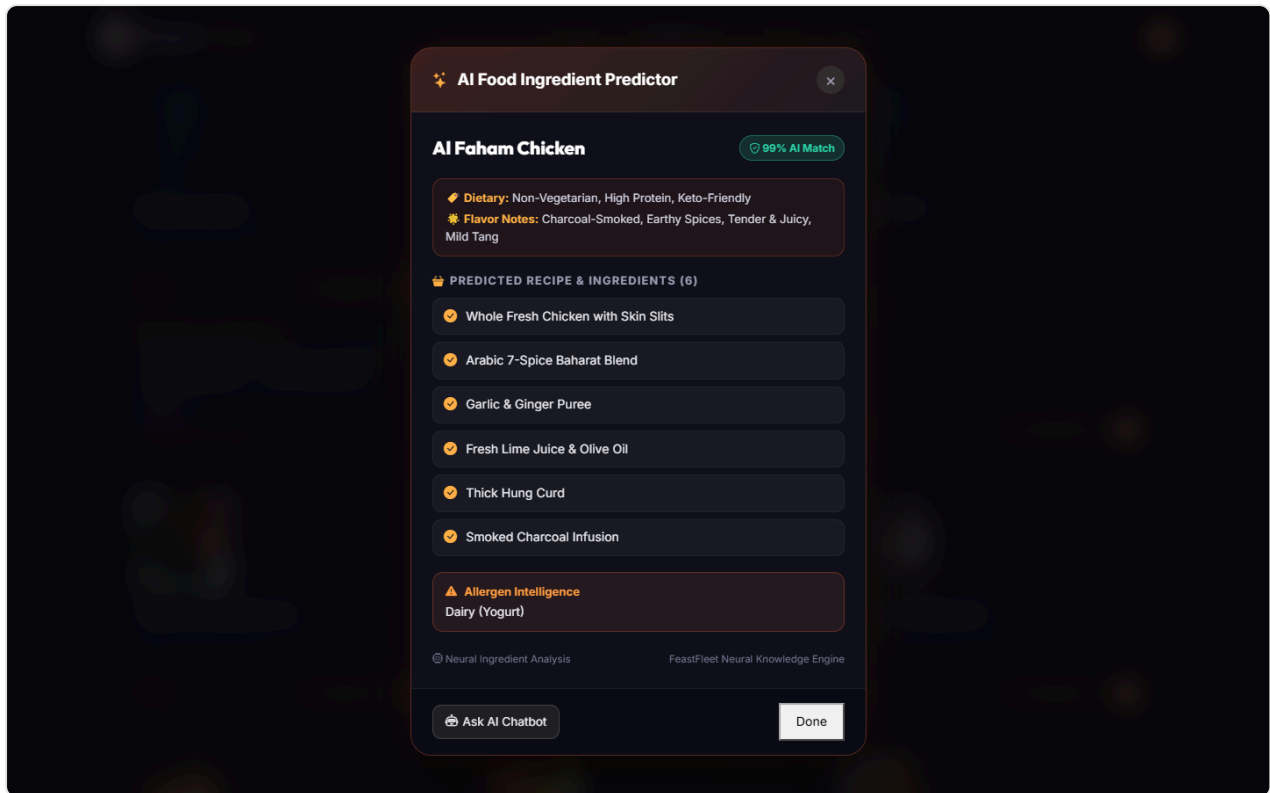


Figure 8. AI Food Ingredient Predictor for Al Faham Chicken.

5.5 Cart and checkout

1. Open **Cart** from the navbar.
2. Change quantity with – / +, or delete an item.
3. Optional: enter a promo code and click **Apply**.
4. Review subtotal, delivery fee (typically ₹49 in the cart summary), and total.
5. Click **Place Order**.

Expected result: cart clears, you are taken to live tracking, and the last order is stored for this browser session.

Items from a second restaurant are allowed only if that restaurant is within **1 km** of restaurants already in the cart. Otherwise a warning toast is shown and the item is not added.

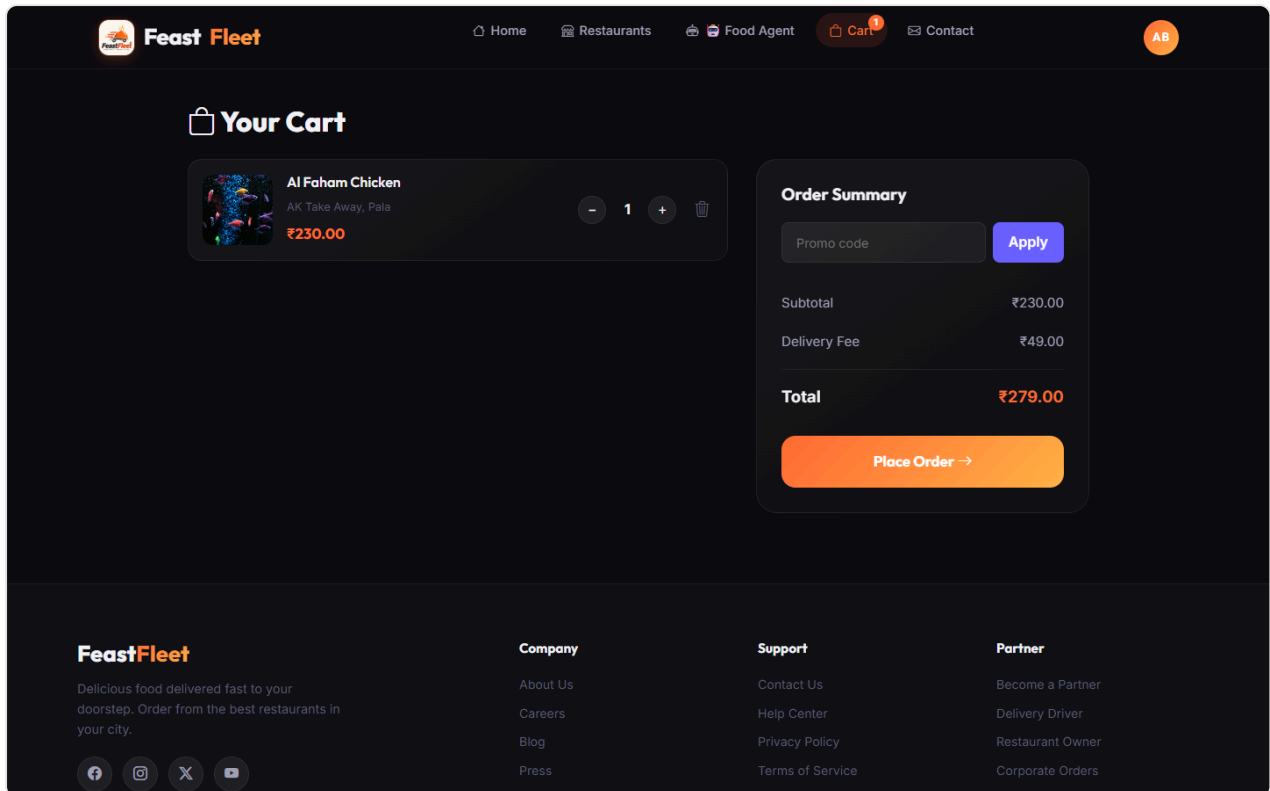


Figure 9. Cart with order summary and Place Order.

5.6 Live order tracking

1. After placing an order, the tracking page opens automatically, or choose **Track Live Order** from the profile menu.
2. Watch stages: Confirmed → Preparing → On The Way → Delivered.
3. The countdown and rider card update in real time. Leaving and returning does not reset the timer to the full duration.

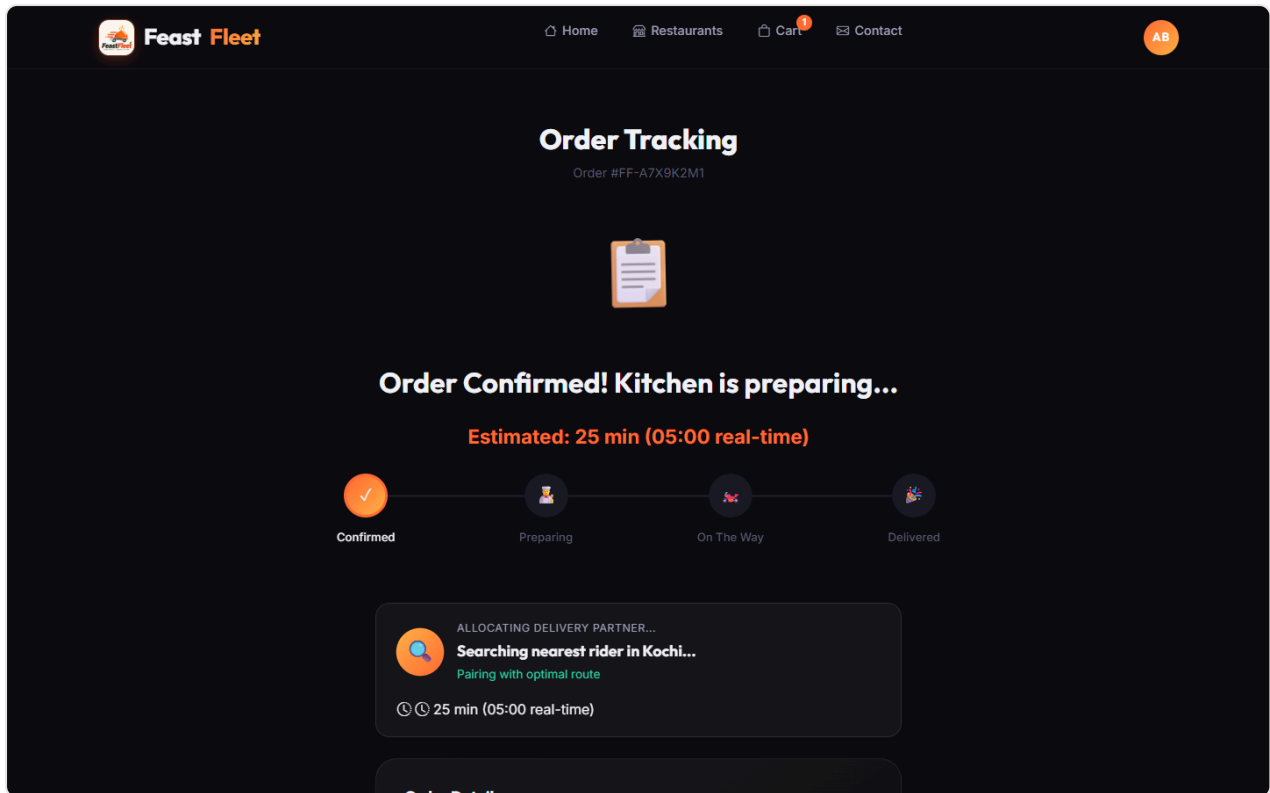


Figure 10. Live tracking timeline and rider allocation.

5.7 Order history

1. Open the profile avatar → **Order History** (`orders.html`).
2. Logged-in users see orders associated with their account when the API is available.

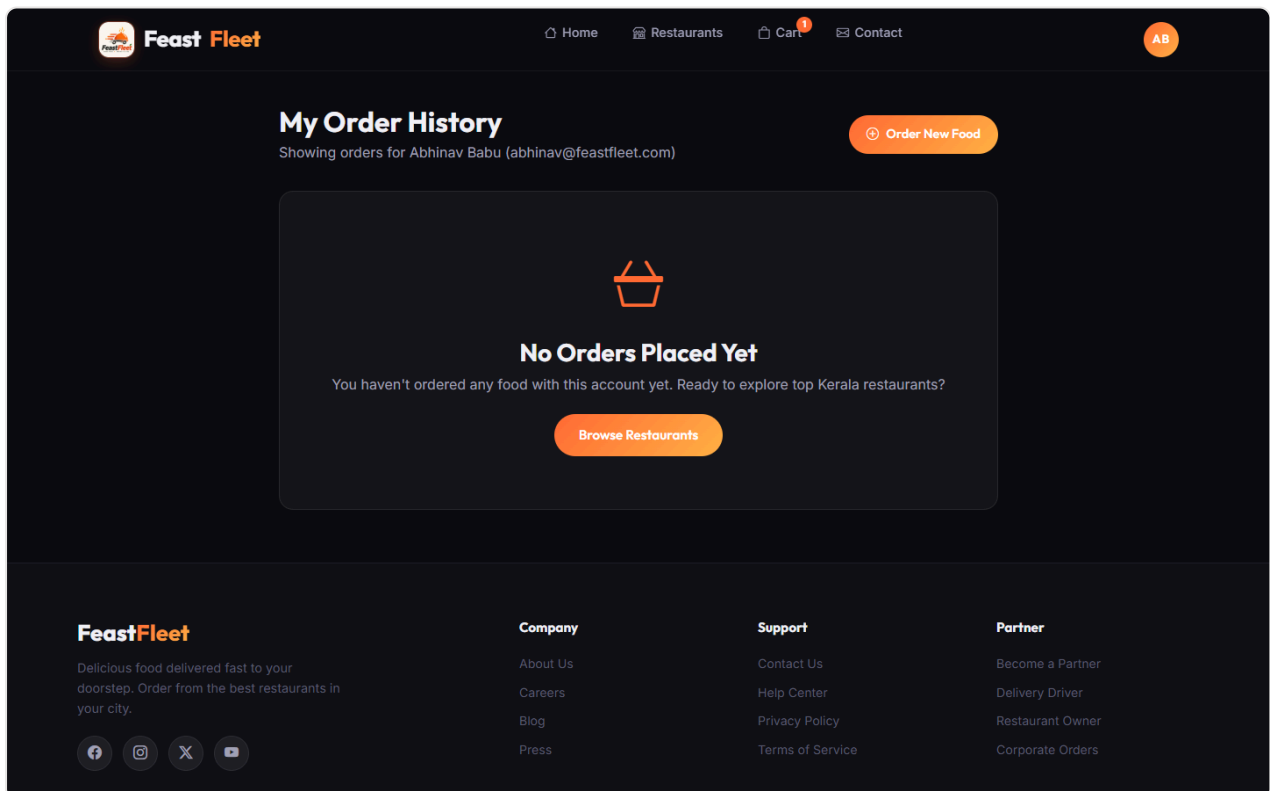
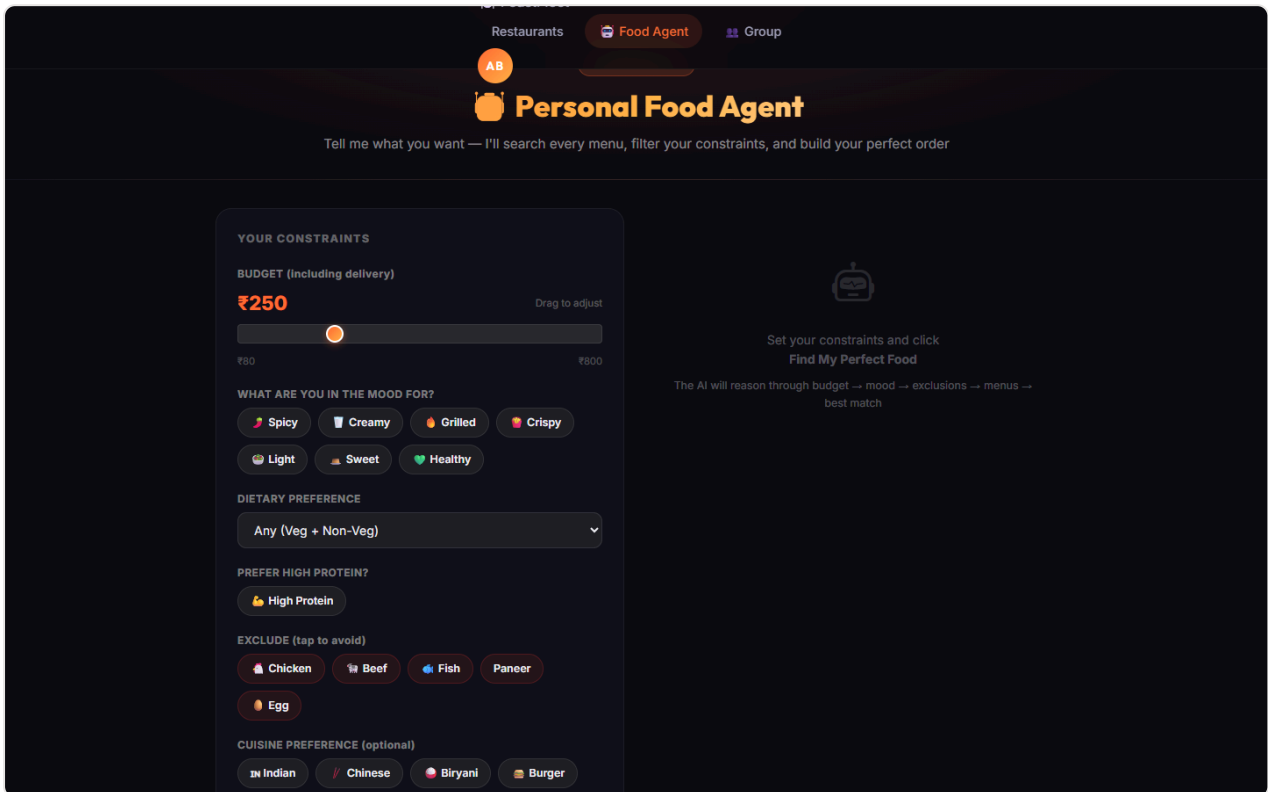


Figure 11. Order history.

5.8 Personal Food Agent

1. From Home, open **Personal Food Agent** (or navbar Food Agent).
2. Set budget, mood tags (spicy, grilled, etc.), diet, exclusions, and cuisine.
3. Run the agent to get a constrained meal suggestion that can be added to the cart.



The screenshot shows the 'Personal Food Agent' interface. At the top, there are navigation tabs: 'Restaurants', 'Food Agent' (selected), and 'Group'. Below the tabs is a header with the title 'Personal Food Agent' and a subtitle 'Tell me what you want — I'll search every menu, filter your constraints, and build your perfect order'. The main content area is divided into two sections. On the left, under the heading 'YOUR CONSTRAINTS', there are several sections: 'BUDGET (including delivery)' with a slider set to ₹250; 'WHAT ARE YOU IN THE MOOD FOR?' with buttons for Spicy, Creamy, Grilled, Crispy, Light, Sweet, and Healthy; 'DIETARY PREFERENCE' with a dropdown menu set to 'Any (Veg + Non-Veg)'; 'PREFER HIGH PROTEIN?' with a button for High Protein; 'EXCLUDE (tap to avoid)' with buttons for Chicken, Beef, Fish, Paneer, and Egg; and 'CUISINE PREFERENCE (optional)' with buttons for Indian, Chinese, Biryani, and Burger. On the right, there is a large button labeled 'Find My Perfect Food' and a text box explaining the AI's reasoning process: 'The AI will reason through budget → mood → exclusions → menus → best match'.

Figure 12. Personal Food Agent constraints panel.

5.9 Budget optimizer

1. Open **Budget Optimizer** from Home.
2. Enter group size, total budget, and meal preferences.
3. Run optimize to get a combination of dishes that fits price and delivery constraints.

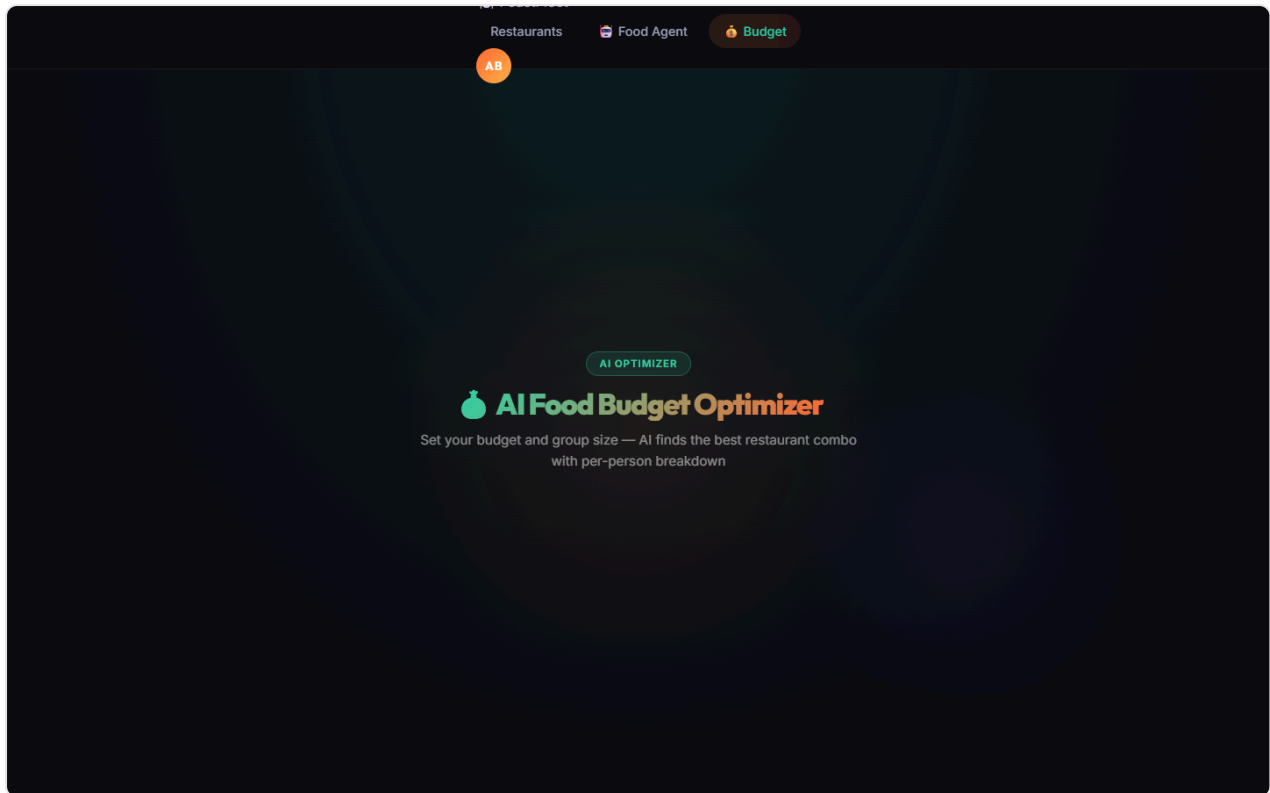


Figure 13. Budget Optimizer.

5.10 Food memory profile

1. Open **Food Memory Profile**.
2. Review patterns built from previous orders in this browser (weekday habits, price band, repetition).
3. Use suggested variations when the profile recommends a change of dish.

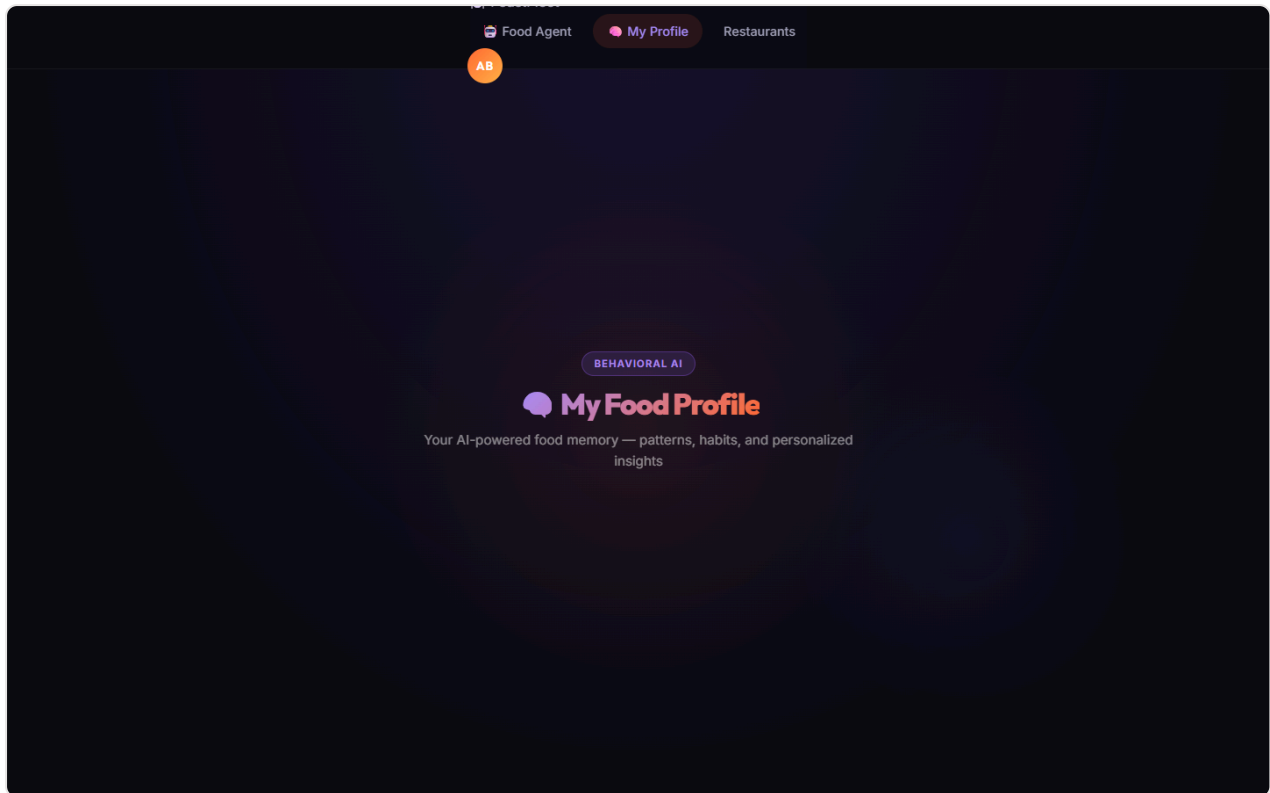


Figure 14. Food Memory Profile.

5.11 Visual food search

1. Open **See Food** → **Find Food**.
2. Upload a food photo or pick a visual category.
3. The system matches nearby menu items and lists the closest dishes to order.

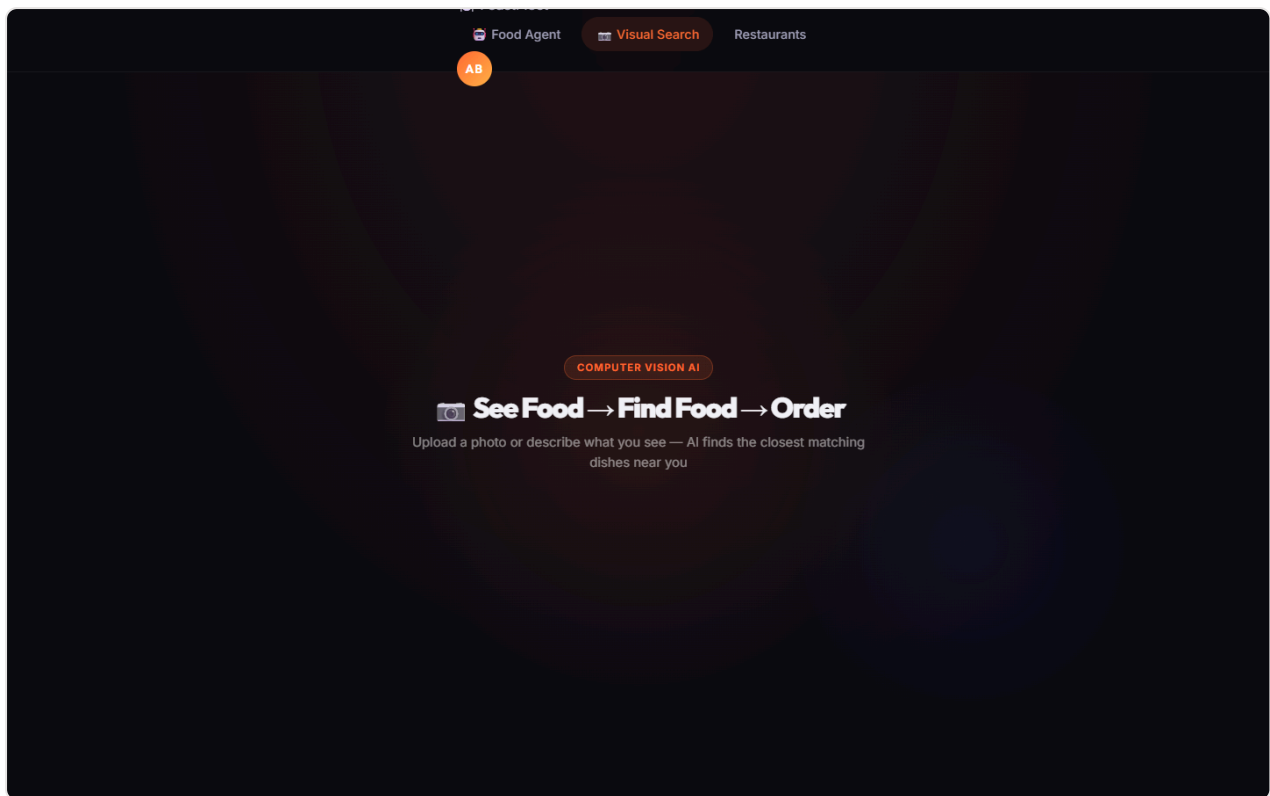


Figure 15. Visual food search.

5.12 AI group order

1. Open **AI Group Order Agent**.
2. Set total group budget and maximum delivery time.
3. For each person, enter name, food preference, diet, per-person budget, and optional exclusions.
4. Add or remove members, then run the agent.
5. **Output:** one recommended restaurant, a dish per person, and a per-person split. You can send the plan to the cart.

RestaurantsOrdersHelp

AB

Group Order Agent

Add your friends' preferences — AI picks the best single restaurant for everyone with optimal dish assignments

Group Settings

TOTAL GROUP BUDGET (₹)

800

MAX DELIVERY TIME

35 minutes

Group Members

Person 1

NAME

Rahul

FOOD PREFERENCE

biryani

DIETARY

Any (Veg + Non-Veg)

BUDGET PER PERSON (₹)

200

EXCLUSIONS (optional)

e.g. no onion, no spicy

Person 2

NAME

Anu

FOOD PREFERENCE

paneer

DIETARY

Vegetarian Only

BUDGET PER PERSON (₹)

150

Remove

Figure 16. Group Order Agent with sample members.

5.13 FeastBot chatbot

1. If the panel is minimized, click the circular FeastBot button (bottom-right).
2. Use quick replies (Track Order, Food Agent, Budget Optimizer, etc.) or type a question.
3. Minimize with the header control when you need the full page.

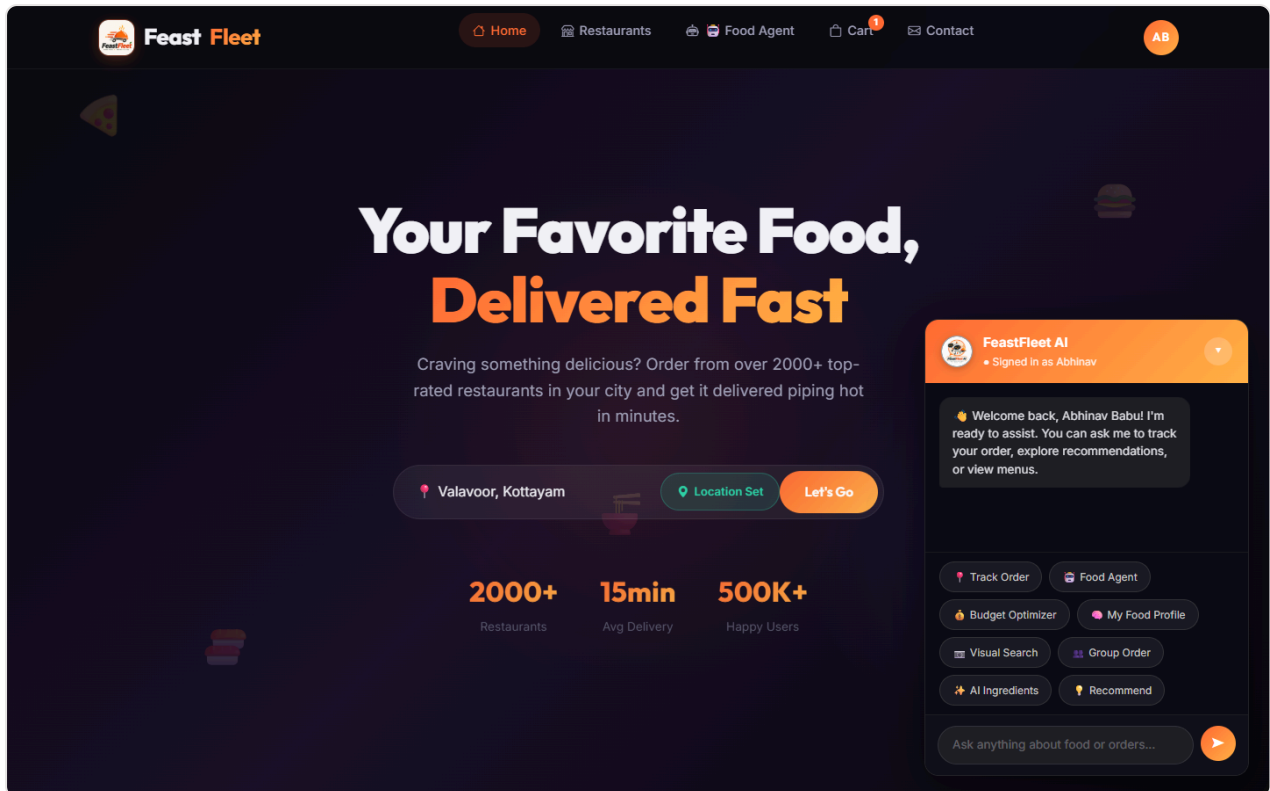


Figure 17. FeastBot assistant on the home page.

5.14 Contact and support

1. Open **Contact**.
2. Fill name, email, subject, and message, then **Send Message**.
3. Use listed email `support@feastfleet.com` , phone `+91 484 123 4567` , or the FAQ accordion on the same page.

Feast Fleet Home Restaurants Cart ¹ Contact AB

Get in Touch

Have questions? We'd love to hear from you.

Send us a Message

Your Name

Your Email

Select Subject

Your Message

Send Message

Email Us
support@feastfleet.com

Call Us
+91 484 123 4567

Visit Us
Infopark, Kochi, Kerala, India

Figure 18. Contact form and support details.

6. Navigation

Primary movement is through the top navbar. On narrow screens, open the hamburger menu to reach the same links.

Control	Goes to
Logo / Home	Storefront (<code>index.html</code>)
Restaurants	Restaurant catalog
Food Agent	Personal Food Agent (when shown)
Cart (with badge)	Cart and checkout
Contact	Support form and FAQ
Sign In	Login / registration
Profile avatar (after login)	Order History, Track Live Order, My Cart, Support, Log Out
Home AI cards	Agent, Budget, Food Profile, Visual Search, Group Order
Restaurant card	That restaurant's menu
Place Order	Tracking page

Control	Goes to
FeastBot quick replies	Matching tool or tracking conversation

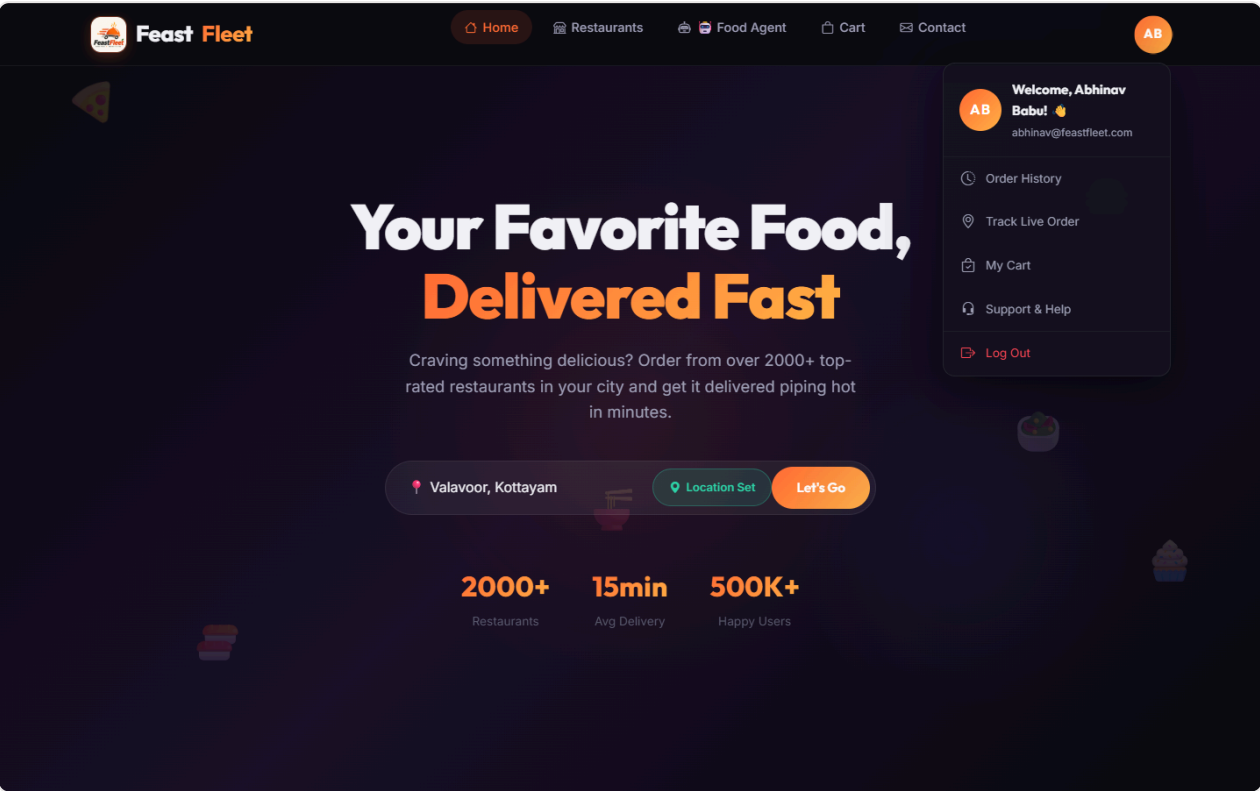


Figure 19. Profile menu used to jump between orders, tracking, cart, support, and logout.

7. Input and Output

Screen	Important inputs	Expected output
Login	Identity + password	Session, avatar, redirect to Home
Sign Up	Name, phone, email, password, 4-digit OTP	New account and signed-in session
Home location	Address or Detect Location	Location label; distance badges on restaurants
Menu	Category filter; + to add; AI Ingredients	Cart badge; ingredient modal
Cart	Qty, delete, promo, Place Order	Updated totals; tracking page
Tracking	(none after order)	Status steps, ETA, rider card
Food Agent	Budget, mood, diet, exclusions	Recommended dishes within constraints
Budget Optimizer	People count, budget, cuisine	Priced combo within budget

Screen	Important inputs	Expected output
Group Order	Members, diets, budgets, max delivery time	One restaurant, assigned dishes, split
Visual Search	Photo or category	Matching menu items
Contact	Name, email, subject, message	Success toast that the message was sent

8. Error Handling / Messages

Message / situation	Cause	What the user should do
"Please fill in all required fields"	Empty identity or password	Complete the form and submit again.
"Please enter your full name / username"	Sign up without a name	Enter full name.
"Invalid credentials"	Wrong email/mobile or password	Recheck details or use demo login.
"User already exists"	Email already registered	Sign In instead of Sign Up.
Password too short (server)	Password under 6 characters	Choose a longer password.
Toast: restaurant is X km from another, must be within 1 km	Multi-restaurant cart rule	Finish the first restaurant's order, or pick a closer second restaurant.
Empty cart: "Your cart is empty"	No items added	Browse restaurants and add dishes.
Place Order does nothing	Cart still empty	Add at least one item.
Group order alert: add at least one member	No member cards	Add a member and run again.
API / network failure	Server not running	Start <code>npm start</code> . Auth may still work in local demo mode.
401 session cleared	Expired or invalid token	Sign in again.

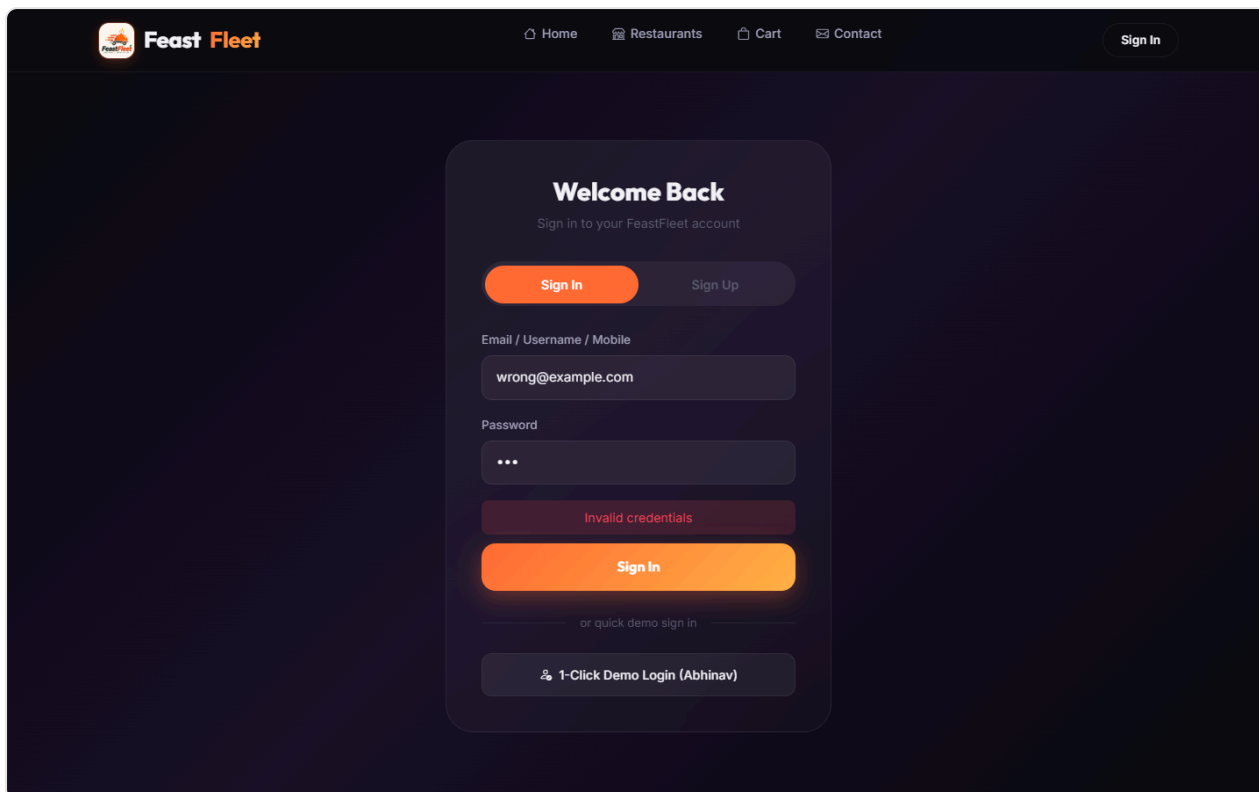


Figure 20. Invalid credentials shown after a failed sign-in.

9. Logout / Exit

1. Click the circular **profile avatar** (initials) at the top right.
2. Click **Log Out** (red item at the bottom of the menu).
3. The token and stored user are removed. The navbar returns to **Sign In** and you are sent to Home.

To leave the application entirely, close the browser tab. Stopping the backend: in the terminal where the server is running, press **Ctrl+C**.

Logging out does not always wipe the cart in localStorage. If you share a computer, also clear the cart or use a private window.

10. Troubleshooting / FAQ

Problem	Solution
Page will not load / connection refused	Confirm <code>npm start</code> is running and visit <code>http://localhost:5000</code> , not a random Live Server port, if the API must work.
Port 5000 already in use	Stop the other process or set environment variable <code>PORT</code> to another value and use that URL.
MongoDB not installed	No action required for a typical demo; in-memory MongoDB starts automatically.

Problem	Solution
Images or Bootstrap look broken	Connect to the internet so CDNs can load, or refresh after the network is up.
Location never changes from Valavoor	This demonstration build is configured to Valavoor, Kottayam for a consistent review.
Cannot add a second restaurant	Respect the 1 km cart radius, or place the first order first.
Tracking timer looks “stuck” after refresh	It continues from elapsed time; it does not restart at 5 minutes.
Chatbot covers the page	Click the minimize (▾) control on the FeastBot header.
How do I track an order?	Place an order, or open profile → Track Live Order, or ask FeastBot “Track my order”.
What payment methods?	Described on Contact FAQ as cash, cards, and Apple Pay (demonstration copy).

11. Appendix — Screenshot Index

All figures are captured from the implemented system at `http://localhost:5000` . Source files live in `docs/screenshots/` .

- 01-splash.png — Launch splash
- 02-home.png — Home
- 03-login.png — Sign In
- 04-register.png — Sign Up
- 05-login-error.png — Invalid credentials
- 06-home-logged-in.png — Authenticated home
- 07-profile-menu.png — Profile / logout menu
- 08-restaurants.png — Restaurant list
- 09-menu-ak.png — AK Take Away menu
- 10-ai-ingredients.png — Ingredient predictor
- 11-cart.png — Cart
- 12-tracking.png — Live tracking
- 13-orders.png — Order history
- 14-ai-agent.png — Food Agent
- 15-budget-optimizer.png — Budget Optimizer
- 16-food-profile.png — Food Memory
- 17-visual-search.png — Visual search
- 18-group-order.png — Group Order Agent

19. 19-contact.png — Contact

20. 20-chatbot.png — FeastBot

End of user manual. Submit this PDF together with the screenshot folder and updated source code as required by the course.